Chapter 3 — Circular and Rotational Motion

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Q.1 — Multiple Choice Questions (with correct option and reason)

3.1. The ratio of angular speed of minute's hand and hour's hand of a watch is:

(a) 1:6 **(b)** 6:1
(c) 1:12 **(d)** 12:1

Answer: (d) 12:1 — Minute hand completes 12 revolutions while hour hand completes 1.

3.2. A body travelling in a circle at constant speed:

(a) has constant velocity **(b)** has an inward radial acceleration
(c) is not accelerated **(d)** has an outward radial acceleration

Answer: (b) has an inward radial acceleration — Direction changes, so acceleration is inward.

3.3. A stone whirled in a vertical circle at constant speed has minimum tension at:

(a) the top of the circle **(b)** halfway down
(c) the bottom of the circle **(d)** anywhere

Answer: (a) the top of the circle — At the top, $T = mv^2/r - mg$.

3.4. Every point of a rotating rigid body has:

(a) same angular velocity **(b)** same linear velocity
(c) same linear acceleration **(d)** same linear distance

Answer: (a) same angular velocity — $v = r\omega$ differs with radius, but ω is common.

3.5. The minimum velocity necessary to put a satellite into orbit is called:

(a) terminal velocity **(b)** critical velocity
(c) artificial velocity **(d)** angular velocity

Answer: (b) critical velocity — Critical or orbital velocity.

3.6. An astronaut orbiting Earth in a large capsule:

(a) is weightless relative to capsule **(b)** is freely falling toward Earth
(c) sees a projected ball move straight **(d)** all of the above

Answer: (d) all of the above — All statements describe free fall in orbit.

3.7. An object makes 10 revolutions in 2 s. Which is true?

(a) period is 2.0 s **(b)** period is 20 s
(c) frequency is 5 Hz **(d)** frequency is 0.2 Hz

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Answer: (c) frequency is 5 Hz — f = 10/2 = 5 Hz.

3.8. A man inside an artificial satellite feels weightless because Earth's attraction:

(a) is zero at pole
(b) is balanced by Moon's attraction
(c) is equal to the centripetal force
(d) is non-effective due to design

Answer: (c) is equal to the centripetal force — Gravity supplies the centripetal force, giving free fall.

3.9. A soda-water bottle swung in a vertical circle has bubbles collecting near the:

(a) bottom
(b) middle
(c) throughout volume
(d) neck

Answer: (d) neck — Light bubbles move toward the centre of rotation, near the neck.

3.10. The moment of inertia of a body depends upon:

(a) mass and its distribution about the axis
(b) volume
(c) kinetic energy
(d) angular momentum

Answer: (a) mass and its distribution about the axis — $I = \Sigma mr^2$.

Q.2 — Short Answer Questions

3.1. State the second law of motion in case of rotation.

The net torque on a rigid body equals the product of its moment of inertia and angular acceleration: $\tau = I\alpha$.

3.2. What is the effect of changing the position of a diver while diving?

By tucking, the diver reduces moment of inertia. With angular momentum conserved, angular velocity increases, so the diver spins faster. Stretching slows the spin.

3.3. How do we get butter from milk using a centrifuge?

Rapid rotation requires a large centripetal force. Denser particles move outward more strongly, separating cream/butter from the lighter milk.

3.4. Mass is a measure of inertia in linear motion. What is its analogue in rotational motion?

Moment of inertia $I = \Sigma mr^2$ is the rotational analogue of mass. It measures resistance to angular acceleration and depends on mass distribution about the axis.

3.5. Why is it harder for a car to take a turn at higher speed?

Required centripetal force is mv^2/r. At higher speed the needed friction is much larger; if friction cannot supply it, the car skids.

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3.6. What are the benefits of double rear tyres on heavy vehicles?

Double tyres increase contact area and load capacity, reduce pressure on the road, improve grip and provide safety if one tyre fails.

3.7. When a car turns left, in which direction do occupants tend to fall?

They tend to fall to the right because their bodies continue in the original straight-line motion due to inertia while the car turns left.

3.8. Why is acceleration of a body moving uniformly in a circle directed toward the centre?

The velocity changes direction toward the inside of the circle. The change in velocity points inward, so acceleration is centripetal.

3.9. How does an astronaut feel weightless while orbiting Earth?

Astronaut and spacecraft fall freely with the same acceleration. There is no normal force between them, so the astronaut feels weightless.

3.10. If angular velocity of particles of a rigid body is constant, is linear velocity also constant?

No. Linear velocity $v = r\omega$ depends on distance from the axis, so particles at different radii have different linear velocities. Also direction changes continuously.

3.11. Why do we tumble when taking a sharp turn at large speed?

The required centripetal force mv^2/r becomes large. If friction cannot provide it, the body continues outward relative to the turn and we tumble.

3.12. What is the time period of a simple pendulum in an artificial satellite?

In orbit the effective gravity is zero, so the restoring force is absent. The pendulum does not oscillate and its period is effectively infinite.

3.13. Is the motion of a satellite in orbit uniform or accelerated?

It is accelerated because the direction of velocity changes continuously, even if speed is constant.

3.14. Why is radian preferred as SI unit over degree?

Radian is a natural ratio of arc to radius and makes angular equations such as $s = r\theta$ and $v = r\omega$ simple without conversion factors.

3.15. In uniform circular motion, what are average velocity and average acceleration for one revolution?

Both are zero because displacement is zero and the velocity returns to its initial value after one complete revolution.

3.16. In a rainstorm with strong wind, what determines the best position to hold an umbrella?

The umbrella should be tilted opposite to the resultant velocity of rain relative to the person, which combines vertical rain velocity and horizontal wind velocity.

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3.17. A ball just supported by a string breaks it when whirled in a vertical circle. Why?

At the bottom the tension must supply both centripetal force and weight: $T = mv^2/r + mg$, which is greater than the weight alone.

3.18. How is centripetal force supplied for (i) a satellite (ii) a car turning on a level road?

(i) Earth's gravitational attraction supplies it. (ii) Static friction between tyres and road supplies it.

Comprehensive / Long Questions

3.1 What is angular momentum? Explain conservation of angular momentum with daily-life examples.

Angular momentum of a rotating body is $L = I\omega$. If no external torque acts, L remains constant. A skater spins faster when arms are pulled in because I decreases and ω increases. Divers and gymnasts use the same principle.

3.2 Show that orbital angular momentum is L = mvr.

For a satellite of mass m moving with speed v at radius r, linear momentum is mv. Since velocity is perpendicular to radius, angular momentum $L = r \times p$ has magnitude mvr.

3.3 Define moment of inertia. Prove τ = Iα for a rotating rigid body.

Moment of inertia $I = \Sigma mr^2$. For each mass element, tangential force $F = ma_t = mr\alpha$. Torque $\tau = rF = mr^2\alpha$. Summing over all elements gives $\tau = (\Sigma mr^2)\alpha = I\alpha$.

3.4 What are artificial satellites? Calculate the minimum time period near Earth's surface.

Artificial satellites are bodies placed in orbit around Earth. For a near-surface orbit, $v = \sqrt{(gR)}$ and $T = 2\pi R/v = 2\pi\sqrt{(R/g)}$. With $R = 6.4 \times 10^6$ m and g = 9.8 m s^{-2}, $T \approx 5.1 \times 10^3$ s ≈ 84 min.

3.5 Define orbital velocity and derive an expression for it.

Orbital velocity is the speed needed for a stable circular orbit. Equating gravity to centripetal force, $GMm/r^2 = mv^2/r$, gives $v = \sqrt{(GM/r)}$.

3.6 Write a note on artificial gravity. Derive the frequency for a rotating spaceship.

In a rotating station, the wall's normal force provides centripetal acceleration and feels like weight. Setting $\omega^2 R = g$ gives $\omega = \sqrt{(g/R)}$, so $f = (1/2\pi)\sqrt{(g/R)}$.

3.7 Prove (i) v = rω and (ii) a = rα.

(i) Arc length $s = r\theta$; differentiating gives $v = r\omega$. (ii) Differentiating $v = r\omega$ gives tangential acceleration $a_t = r\alpha$.

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Numerical Problems (fully worked)

3.1 A laser beam spreads to diameter 2.50 m at the Moon, distance 3.8 × 10⁸ m. Find divergence angle.

Given: diameter = 2.50 m, distance = 3.8×10^8 m

Formula: θ ≈ diameter/distance

θ = $2.50/(3.8 \times 10^8) \approx 6.6 \times 10^{-9}$ rad.

Answer: 6.6 × 10⁻⁹ rad

3.2 A car moves at 108 km h⁻¹. Wheel diameter 60 cm. Find angular speed in rad s⁻¹ and rev s⁻¹.

Given: v = 30 m s^{-1}, r = 0.30 m

Formula: ω = v/r, f = ω/2π

ω = 30/0.30 = 100 rad s^{-1}.

f = 100/(2π) ≈ 16 rev s^{-1}.

Answer: 100 rad s⁻¹, 16 rev s⁻¹

3.3 A motor runs at 1800 rev min⁻¹ and stops in 20 s with uniform retardation. Find revolutions before stopping.

Given: f_0 = 30 rev s^{-1}, f = 0, t = 20 s

Formula: N = average frequency × time

Average frequency = (30 + 0)/2 = 15 rev s^{-1}.

N = 15 × 20 = 300 rev.

Answer: 300 rev

3.4 A string 0.6 m long withstands 35.6 N. Find maximum speed of a 0.5 kg stone whirled in a vertical circle.

Given: r = 0.6 m, T_max = 35.6 N, m = 0.5 kg

Formula: At bottom: T − mg = mv²/r

v = √[(T − mg)r/m] = √[(35.6 − 4.9)×0.6/0.5] ≈ 6.1 m s^{-1}.

Answer: ≈ 6.1 m s⁻¹

The book prints 5.5 m s⁻¹; using T − mg at the bottom gives about 6.1 m s⁻¹.

3.5 A flywheel at 2100 rev min⁻¹ stops in 3 min. I = 36 kg m². Find torque.

Given: ω_0 = 2100×2π/60 ≈ 220 rad s^{-1}, t = 180 s, I = 36 kg m²

Formula: τ = Iα, α = −ω_0/t

α = −220/180 ≈ −1.22 rad s^{-2}.

|τ| = 36 × 1.22 ≈ 44 N m.

Answer: 44 N m

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3.6 Find moment of inertia of a 200 kg sphere of diameter 60 cm.

Given: M = 200 kg, R = 0.30 m

Formula: $I = (2/5)MR^2$

$I = 0.4 \times 200 \times 0.09 = 7.2$ kg m².

Answer: 7.2 kg m²

3.7 A satellite orbits at altitude 200 km. Earth radius 6400 km. Find orbital speed.

Given: $r = 6.6 \times 10^6$ m, g = 9.8 m s^{-2}, $R = 6.4 \times 10^6$ m

Formula: $v = \sqrt{(gR^2/r)}$

$v = \sqrt{(9.8 \times (6.4\times10^6)^2/(6.6\times10^6))} \approx 7.77 \times 10^3$ m s^{-1}.

Answer: 7.77 km s⁻¹

3.8 A space station radius 20 m rotates at 0.5 rad s⁻¹. Find artificial gravity.

Given: R = 20 m, ω = 0.5 rad s^{-1}

Formula: $a = \omega^2 R$

$a = 0.25 \times 20 = 5$ m s^{-2}.

Answer: 5 m s⁻²

3.9 A bicycle wheel has angular momentum 10 kg m² s⁻¹ and angular velocity 2 rad s⁻¹. Find I.

Given: L = 10, ω = 2

Formula: $I = L/\omega$

$I = 10/2 = 5$ kg m².

Answer: 5 kg m²

3.10 A diver has I = 18 kg m² straight, then tucks to 3.6 kg m² and makes 2 rotations in 1.0 s. How many revolutions in 1.5 s if not tucked?

Given: $I_1 = 18$, $I_2 = 3.6$, $f_2 = 2$ rev s^{-1}

Formula: $I_1\omega_1 = I_2\omega_2$

$f_1 = (I_2/I_1)f_2 = (3.6/18)\times2 = 0.4$ rev s^{-1}.

In 1.5 s: $N = 0.4 \times 1.5 = 0.6$ rev.

Answer: 0.6 rev

Textbook checks flagged in this chapter

• The short-question numbering in the book appears to repeat/continue irregularly around 3.10; the questions are answered in book order.

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• Numerical 3.4: book prints 5.5 m s^{-1}; the bottom-of-circle tension equation gives about 6.1 m s^{-1}.

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